A PROOF OF A CONJECTURE OF BĂLĂNESCU AND CIMPOEAŞ ON THE HILBERT DEPTH OF INTERSECTIONS OF MONOMIAL PRIMES

Abstract. Let $S = K[x_1, \dots, x_N]$, and let I be the intersection of monomial prime ideals generated by pairwise disjoint blocks of variables of sizes $n_1, \dots, n_r$, where $N = n_1 + \cdots + n_r$. Bălănescu and Cimpoeaş proved

$$\operatorname{hdepth}(I) \le \left\lfloor \frac{N+r}{2} \right\rfloor$$

and conjectured equality. We prove the matching lower bound, thereby resolving their Conjecture 3.4. The proof combines Uliczka's criterion with a coefficientwise-positivity lemma for centered shifts of palindromic polynomials.

1. Introduction

Let K be a field. Partition the variables of $S = K[x_1, \dots, x_N]$ into nonempty blocks $B_1, \dots, B_r$ of sizes $n_1, \dots, n_r$, and set

$$P_i = (x_j : j \in B_i), \qquad I = P_1 \cap \cdots \cap P_r, \qquad N = n_1 + \cdots + n_r.$$

Bălănescu and Cimpoeaş proved the upper bound

$$\operatorname{hdepth}(I) \le \left\lfloor \frac{N+r}{2} \right\rfloor$$

and obtained equality for $r = 2$ and whenever at most one of the n_i is even [1, Theorems 2.9 and 3.3]. They proposed equality in general as Conjecture 3.4. Our contribution is the reverse inequality for arbitrary positive block sizes.

Theorem 1. *With the notation above,*

$$\operatorname{hdepth}(I) = \left\lfloor \frac{N+r}{2} \right\rfloor .$$

In particular, Conjecture 3.4 of Bălănescu and Cimpoeaş is true.

2. Proof

A formal power series is called nonnegative if all its coefficients are nonnegative.

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Lemma 2. *Let*

$$Q(u) = \sum_{j=0}^{D} q_j u^j, \qquad q_j = q_{D-j} \geq 0,$$

and put $\delta = \lceil D/2 \rceil$*. Then*

$$(1-t)^{-\delta} Q(1-t)$$

is a nonnegative formal power series.

Proof. We first note that, for integers $b \geq a \geq 0$,

$$(1-t)^a + (1-t)^{-b} \tag{1}$$

is nonnegative. The case $b = 0$ is immediate. If $b \geq 1$, then for every $m \geq 0$ its coefficient of t^m is

$$(-1)^m \binom{a}{m} + \binom{b+m-1}{m}.$$

This is nonnegative when m is even or $m > a$. If m is odd and $1 \leq m \leq a$, then $b + m - 1 \geq a$, so

$$\binom{b+m-1}{m} \geq \binom{a}{m}.$$

Thus (1) is nonnegative.

If $D = 2\delta$, palindromicity gives

$$u^{-\delta} Q(u) = q_\delta + \sum_{e=1}^{\delta} q_{\delta+e}(u^e + u^{-e}).$$

If $D = 2\delta - 1$, it gives

$$u^{-\delta} Q(u) = \sum_{e=0}^{\delta-1} q_{\delta+e}(u^e + u^{-e-1}).$$

Substituting $u = 1 - t$ and applying (1) with $(a, b) = (e, e)$ in the first case and $(a, b) = (e, e+1)$ in the second proves the claim. $\square$

For a finitely generated graded S-module M, write

$$H_M(t) = \sum_{k \geq 0} \dim_K(M_k) t^k.$$

Uliczka's criterion states that

$$\operatorname{hdepth}(M) = \max\{d \in \mathbb{Z}_{\geq 0} : (1-t)^d H_M(t) \text{ is nonnegative}\} \tag{2}$$

[3, Theorem 3.2]. Bălănescu and Cimpoeaş invoke this criterion in the squarefree reformulation [2, Theorem 2.4], which is derived from it; we use it directly.

Proof of Theorem 1. A monomial lies in I precisely when its support meets every block. Consequently,

$$\begin{aligned} H_I(t) &= \prod_{i=1}^{r} \left((1-t)^{-n_i} - 1\right) \\ &= \frac{t^r Q(1-t)}{(1-t)^N}, \end{aligned}$$

where

$$Q(u) = \prod_{i=1}^{r} [n_i]_u, \qquad [n]_u = 1 + u + \cdots + u^{n-1}.$$

The polynomial Q has nonnegative coefficients, is palindromic, and has degree

$$D = \sum_{i=1}^{r} (n_i - 1) = N - r.$$

Set

$$q = \left\lfloor \frac{N+r}{2} \right\rfloor, \qquad \delta = N - q = \left\lceil \frac{D}{2} \right\rceil.$$

Then

$$(1-t)^q H_I(t) = t^r (1-t)^{-\delta} Q(1-t),$$

which is nonnegative by Lemma 2. Hence (2) yields $\mathrm{hdepth}(I) \geq q$.

The reverse inequality was proved in [1, Theorem 3.3]. For completeness, it also follows from the first coefficient after the initial term. If $d > q$, then

$$\begin{aligned} [t^{r+1}](1-t)^d H_I(t) &= (N-d)Q(1) - Q'(1) \\ &= Q(1)\left(\frac{N+r}{2} - d\right) < 0. \end{aligned}$$

Indeed, palindromicity of degree D gives $Q'(1) = \frac{D}{2} Q(1)$, and $Q(1) = \prod_i n_i > 0$. Thus no $d > q$ is admissible in (2), and $\mathrm{hdepth}(I) = q$. $\square$

Status and scope. Theorem 1 establishes the equality proposed as Conjecture 3.4 of [1] for an arbitrary number of blocks and arbitrary positive block sizes. Only the lower bound is proved here; the upper bound is due to Bălănescu and Cimpoeaş [1, Theorem 3.3], and the alternative derivation given above is included for completeness rather than as a new result. The equality is recorded as a conjecture in the published version of [1] and is still stated as a conjecture in the most recent arXiv revision of that paper (v6, 14 June 2026); we have found no proof of the general case in the literature, but our search is not exhaustive, and the statement is offered subject to external prior-art review. Nothing here bears on the separate conjecture of [1] approximating $\mathrm{hdepth}(S/I)$, which is untouched. The Stanley depth of I is likewise not determined here: the argument concerns Hilbert depth only, and we make no claim about $\mathrm{sdepth}(I)$.

References

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